SolarShare — Implementation Plan

Version: 1.0 | **Companion to:** PRD, TRD, Backend Schema

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1. Phased Roadmap

Phase 0 — Hackathon MVP (Week 0–1)

Goal: Working clickable demo proving the core trade loop.

- Build the 3 core screens (list surplus, browse/buy, ledger history) per the UI/UX Brief
- Simulate smart meter data (no real DISCOM integration)
- Generate tamper-evident transaction IDs via client-side SHA-256 hashing (no real blockchain needed)
- Deliverable: live demo + pitch deck + this documentation set

Phase 1 — Single-Society Pilot (Month 1–3)

Goal: Validate real behavior with a real (small) user base.

- Partner with **one housing society** within a DERC/UPERC pilot zone (Delhi or western UP, per the live IES pilot geography)
- Replace simulated data with manual/CSV-based real meter readings (bridge step before full AMISP integration)
- Onboard 50–200 households via the RWA/society admin, WhatsApp-first
- Track: trades completed, average seller premium, average buyer discount, engagement rate

Phase 2 — Technical Integration (Month 3–6)

Goal: Move from manual bridge to real infrastructure integration.

- Register SolarShare as a **Beckn Application Provider (BAP)** on the UEI network via REC Limited's onboarding pathway
- Integrate real DISCOM AMISP/Head-End System meter feed (DLMS/COSEM)
- Integrate RBI-compliant escrow payment gateway (replacing mock wallet)
- Migrate trust layer from simple hashing to a permissioned blockchain (Hyperledger Fabric) if

trade volume justifies it

Phase 3 — Multi-Society Expansion (Month 6–12)

Goal: Scale within proven, regulator-approved geography before going elsewhere.

- Expand across additional societies within BRPL, TPDDL, and PVVNL service areas
- Introduce AI-based demand/surplus forecasting once sufficient historical data exists
- Launch carbon-credit tracking layer as a second revenue stream
- Begin regulatory engagement in 1–2 additional states (e.g., Maharashtra, Gujarat) for future pilot applications

Phase 4 — Platform Maturity (Year 2+)

- Explore Virtual Power Plant (VPP) partnerships for users who independently adopt battery storage
- Explore EV-charging integration, mirroring how existing TPPs extend from one energy use case to another
- Evaluate interstate trading once CERC-level clearance pathways mature

2. Timeline Overview

Phase	Duration	Key Milestone
0 — Hackathon MVP	1 week	Working demo
1 — Single-society pilot	Months 1–3	50+ active trading users
2 — Technical integration	Months 3–6	Live BAP registration + real meter data
3 — Multi-society expansion	Months 6–12	5+ societies, positive unit economics
4 — Platform maturity	Year 2+	VPP/EV exploration, multi-state presence

3. Team Roles Needed

Role	Phase Needed	Notes
Frontend developer	Phase 0+	React Native / Flutter

Role	Phase Needed	Notes
Backend developer	Phase 0+	Node.js or Python; owns schema from Backend Schema doc
UI/UX design	Phase 0+	Can be a shared role initially per the UI/UX Brief
Regulatory/compliance lead	Phase 1+	Tracks DERC/UPERC pilot terms, DPDP Act compliance
Community/onboarding lead	Phase 1+	Owns RWA partnership and WhatsApp onboarding

Role	Phase Needed	Notes
DevOps	Phase 2+	Owns AMISP/UEI integration reliability

4. Cost Estimates

Item	Phase 0 (Hackathon)	Phase 1 (Pilot)
App/platform development	₹0 (student team)	₹0–1.5 lakh if partial outsourcing needed
Hosting/cloud	₹0 (free tier)	₹0–3,000/month
WhatsApp Business API	₹0	₹2,000–10,000/month (scales with users)
Company registration (if formalized)	—	₹6,000–15,000

Item	Phase 0 (Hackathon)	Phase 1 (Pilot)
Legal/compliance basics	—	₹5,000–20,000
BAP/TPP registration process	—	Primarily time cost, not direct fee, per REC's published onboarding pathway

(Figures are planning estimates based on typical early-stage costs discussed with the team; not a guarantee of actual pricing, which should be confirmed directly with each vendor/authority at the time of implementation.)

5. Key Dependencies & Milestones

- **Regulatory dependency:** Full Phase 2 integration depends on the DERC/UPERC pilot (currently a 6-month pilot from February 2026) either extending or converting into a standing regulatory framework. This is tracked, not controlled, by the team.

- **Society partnership dependency:** Phase 1 cannot begin without a willing RWA/society partner already possessing net-metered solar households — target societies should be pre-screened for existing PM Surya Ghar adoption.
- **UEI onboarding dependency:** BAP registration timeline depends on REC Limited's process, which is new and evolving as of 2026 — plan for schedule flexibility here.

6. Risk Checkpoints

Checkpoint	Trigger to Reassess
End of Phase 1	If trade completion rate is low, revisit matching/pricing logic before scaling
End of Phase 2	If BAP registration stalls, continue operating Phase 1's manual-bridge model rather than blocking on it
End of Phase 3	If unit economics (transaction fee revenue vs. operating cost) aren't trending positive, reassess revenue mix (SaaS/data/carbon credits) before further expansion

7. Definition of Done (per phase)

- **Phase 0:** Judges/mentors can complete a full mock trade in the live demo without guidance.
- **Phase 1:** At least one real household has both listed and sold surplus energy, and at least one real household has bought it, with the trade reflected in an internal report (even if not yet in an official DISCOM bill).
- **Phase 2:** A trade completes end-to-end through the real UEI/Beckn flow with real settlement.